

Support Vector (SVM)

K-Means

# Using Gini-Index to create Decision Tree

Date \_\_\_\_\_  
Page \_\_\_\_\_

Dataset -

Inputs

target

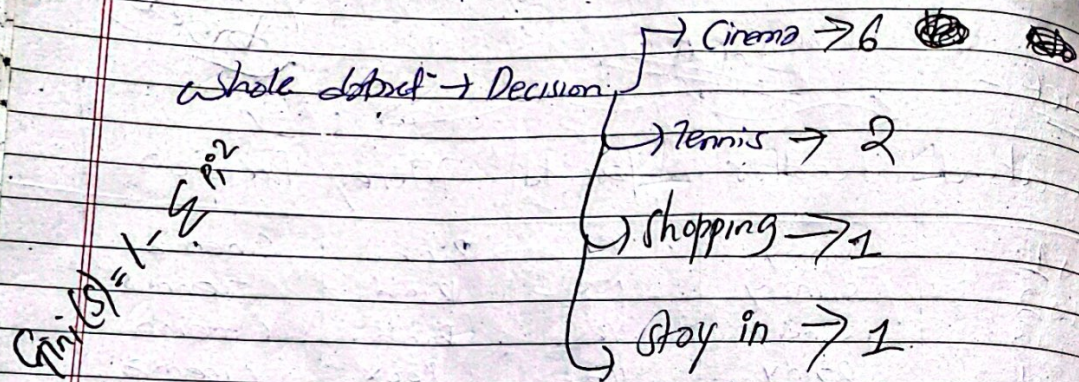
| Weekend | Weather | Parents | Money | Decision          |
|---------|---------|---------|-------|-------------------|
| w1      | Sunny   | Yes     | Rich  | Cinema            |
| w2      | Sunny   | No      | Rich  | Tennis            |
| w3      | Windy   | Yes     | Rich  | Cinema            |
| w4      | Rainy   | Yes     | Poor  | Cinema            |
| w5      | Rainy   | No      | Rich  | Stay In           |
| w6      | Rainy   | Yes     | Poor  | Cinema            |
| w7      | Windy   | No      | Poor  | Shopping (Cinema) |
| w8      | Windy   | No      | Rich  | Shopping          |
| w9      | Windy   | Yes     | Rich  | Cinema            |
| w10     | Sunny   | No      | Rich  | Tennis            |

$\min^2$  gini index  $\Leftrightarrow \max^2$  information gain

And we select that node.



1) Compute the Gini index for the overall collection of training examples.



$$Gini(S) = 1 - \left[ \left(\frac{6}{10}\right)^2 + \left(\frac{2}{10}\right)^2 + \left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right)^2 \right]$$

$$= 0.68$$

2) Calculate gini for every attribute  
root node will have low gini.

for attribute money,

values(money)  $\rightarrow$

|      | Cinema | Tennis | Shopping | Stay in |
|------|--------|--------|----------|---------|
| rich | 7      | 3      | 2        | 1       |
| poor | 3      | 3      | 0        | 0       |

$$Gini(S_{poor}) = 1 - \left( \left(\frac{3}{3}\right)^2 + 0^2 + 0^2 + 0^2 \right)$$

$\stackrel{\text{pretty}}{=} 0$  (one sided)

$$Gini(S_{rich}) = 1 - \left( \left(\frac{2}{9}\right)^2 + \left(\frac{2}{9}\right)^2 + \left(\frac{1}{9}\right)^2 + \left(\frac{1}{9}\right)^2 \right)$$

Imp  $Gini\ Index = 0.694$

Weighted Average (Money)  $= \sum_{VE(poor, rich)} Gini(S_v) * \frac{|S_v|}{S}$

$$= 0 * \frac{3}{10} + 0.694 * \frac{7}{10}$$

$$= 0.486$$



Again,

for attribute parents

|               |   | Cinema | Tennis | PopIn | Shopping |
|---------------|---|--------|--------|-------|----------|
| parents → Yes | 5 | 5      | 0      | 0     | 0        |
| No            | 5 | 2      | 2      | 1     | 1        |

$$Gini(S_{yes}) = 1 - \left( \left( \frac{5}{5} \right)^2 + 0^2 + 0^2 + 0^2 \right)$$

Directly = 0 (only one class on the side)

$$Gini(S_{no}) = 1 - \left( \left( \frac{1}{5} \right)^2 + \left( \frac{2}{5} \right)^2 + \left( \frac{1}{5} \right)^2 + \left( \frac{1}{5} \right)^2 \right)$$

Gini Index = 0.72

$$\text{Weighted Average} = \sum_{v \in \{yes, no\}} Gini(S_v) * \frac{|S_v|}{|S_{parents}|}$$

$$= 0 * \frac{5}{10} + 0.72 * \frac{5}{10}$$

$$= 0.36$$

Again,

for attribute weather,

|                 |   | Cinema | Tennis | PopIn | Shopping |
|-----------------|---|--------|--------|-------|----------|
| weather → sunny | 3 | 1      | 2      | 0     | 0        |
| windy           | 4 | 2      | 0      | 0     | 2        |
| rainy           | 3 | 2      | 0      | 1     | 0        |

$$Gini(S_{sunny}) = 1 - \left( \left( \frac{1}{3} \right)^2 + \left( \frac{2}{3} \right)^2 + 0^2 + 0^2 \right)$$

$$= 0.444$$

$$Gini(S_{windy}) = 1 - \left( \left( \frac{3}{4} \right)^2 + 0^2 + 0^2 + \left( \frac{1}{4} \right)^2 \right)$$

$$= 0.375$$

$$Gini(S_{rainy}) = 1 - \left( \left( \frac{2}{3} \right)^2 + \left( \frac{1}{3} \right)^2 \right)$$

$$= 0.444$$

Gini Index

$$\text{Weighted Average} = \sum_{v \in \{sunny, windy, rainy\}} Gini(S_v) * \frac{|S_v|}{|S_{weather}|}$$

$$= 0.444 * \frac{3}{10} + 0.375 * \frac{4}{10} + 0.444 * \frac{3}{10}$$

$$= 0.417$$



weighted average

Date \_\_\_\_\_  
Page \_\_\_\_\_

for weather, ↓

Gini Index: 0.416

for parents,

Gini Index: 0.36

for money,

Gini Index: 0.486

min<sup>m</sup> Gini Index,  
So its our root  
node.

parents

Yes

No

| Weekend | weather | Parents | Money | Decision |
|---------|---------|---------|-------|----------|
| w1      | Sunny   | Yes     | Rich  | Cinema   |
| w3      | Windy   | Yes     | Rich  | Cinema   |
| w4      | Rainy   | Yes     | Poor  | Cinema   |
| w6      | Rainy   | Yes     | Poor  | Cinema   |
| w9      | Windy   | Yes     | Rich  | Cinema   |

so directly  
cinema  
←

| Weekend | Weather | Parents | Money | Decision |
|---------|---------|---------|-------|----------|
| w2      | Sunny   | No      | Rich  | Temple   |
| w5      | Rainy   | No      | Rich  | Shopping |
| w7      | Windy   | No      | Poor  | Cinema   |
| w8      | Windy   | No      | Rich  | Shopping |
| w10     | Sunny   | No      | Rich  | Temple   |

only for (No) dataset

Date \_\_\_\_\_  
Page \_\_\_\_\_

for attribute weather,

|         |   |        |        |          |
|---------|---|--------|--------|----------|
|         |   | Cinema | Temple | Shopping |
| → sunny | 2 | 0      | 2      | 0        |
| → rainy | 1 | 0      | 0      | 1        |
| → windy | 2 | 1      | 0      | 1        |

$$Gini(S_{sunny}) = 0 \quad (\text{one sided})$$

Directly

$$Gini(S_{rainy}) = \text{Directly } 0 \quad (\text{one sided})$$

$$Gini(S_{windy}) = 1 \quad (\text{Both Equal})$$

Directly,  $1 - \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right)$   
= 0.5

(dataset)

Decision

→ Temple 2

→ Shopping 1

→ Cinema 1

→ Shopping 1

$$Gini \text{ of Dataset} = 1 - \left( \left(\frac{2}{5}\right)^2 + \left(\frac{1}{5}\right)^2 \times 3 \right)$$

= 0.72



Gini Index (Sunny, weather)

Weighted Average =  $\sum \frac{|S_v|}{|S|} * \text{Gain}(S_v)$   
 (Parent = No / weather)  $\forall v \in \{\text{sunny, rainy, windy}\}$

$$= \frac{2}{5} * 0 + \frac{1}{5} * 0 + \frac{2}{5} * 0.5$$

$$= 0.2$$

Again,

for attribute money,

|             | Rich | 4 | 2 | 1 | 1 |
|-------------|------|---|---|---|---|
| Yes → money | 1    | 1 | 0 | 0 | 0 |
| No → money  | 1    | 1 | 0 | 0 | 0 |

$$\text{Gain}(S_{\text{rich}}) = 1 - \left( \left( \frac{1}{4} \right)^2 + \left( \frac{2}{4} \right)^2 + \left( \frac{1}{4} \right)^2 + \left( \frac{1}{4} \right)^2 \right)$$

$$= 0.5$$

$$\text{Gain}(S_{\text{poor}}) = 1 - \text{Diedly (one sided)}$$

Gini Index (Sunny, money)

Weighted Average =  $\sum \frac{|S_v|}{|S_{\text{sunny}}|} * \text{Gain}(S_v)$   
 (Parent = No / Money)  $\forall v \in \{\text{rich, poor}\}$

$$= 0.5 * \frac{4}{5} + 0$$

$$= 0.4$$

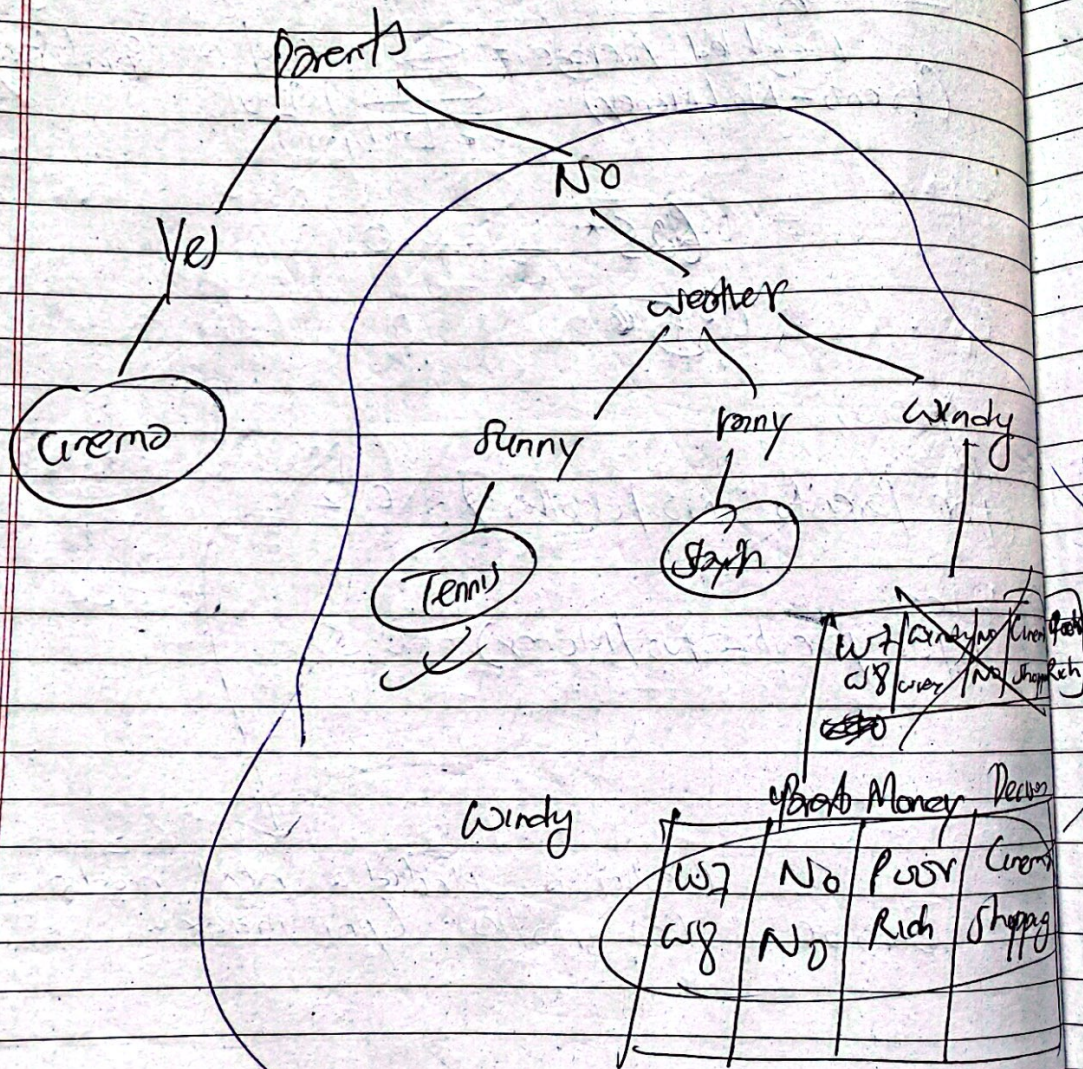
Now,

Gini Index (for Parents = No / weather) = 0.2

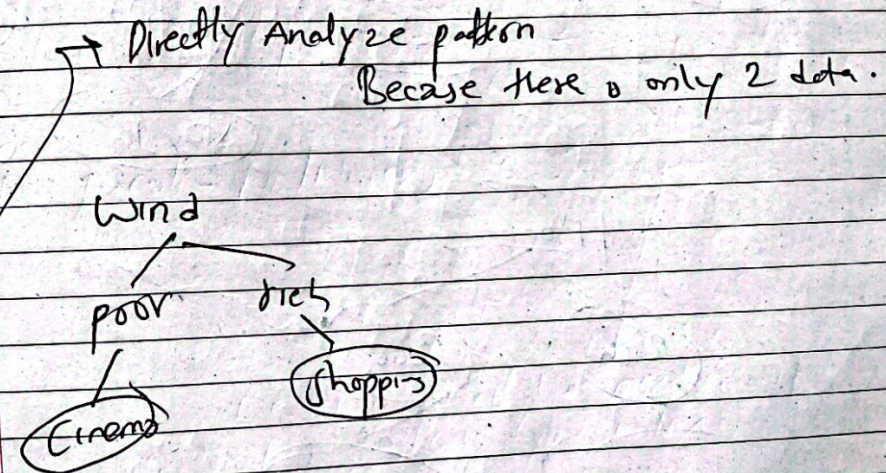
Gini Index (for parent = No / Money) = 0.5

Weather is selected a next node because it has smallest Gini index.





only ~~windy~~ data





So, final Decision Tree,

Date \_\_\_\_\_  
Page \_\_\_\_\_

